Akshat Kankani

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Education

Manipal Institute of Technology — Bengaluru, KA

B.Tech in Computer Science (Artificial Intelligence) — CGPA: 8.11 — *July 2023 – May 2027*

Coursework: *Data Structures & Algorithms, Object-Oriented Programming, DevOps & Cloud, Database Management Systems, Operating Systems, Computer Architecture, Computer Networks, Machine Learning, Generative AI*

Projects

Chronos | *Python, FastAPI, Prometheus, Grafana, GitHub Actions* — GitHub

- Built a **Kubernetes-style orchestrator** in Python with a **versioned CAS store, filter/score scheduler, level-triggered reconciliation** and **rolling updates**.
- Designed **self-fencing nodes** timed from heartbeat send time, guaranteeing **no duplicate replicas** under pause, crash and partition, with failed-node workloads **replaced in 5.3 s**.
- Implemented a **write-ahead log with group commit and snapshots**, raising durable writes **11× (300 → 3,393/s)** and cutting **1M-record recovery from 124 s to 0.22 s**.
- Profiled the scheduler and found **85% of time spent deep-copying cached nodes**; a **read-only cache view** made it **2.8–3.9× faster (221 pods/s at 100 nodes)**.

Strata | *C++, POSIX APIs (file I/O, fsync, signals), Multithreading, Make - Team Project (2 Members)* — GitHub

- Built a C++ **time-series storage engine** with **write-ahead logging, crash recovery and LSM-style compaction.**
- Implemented **Gorilla-based compression** using delta-of-delta and XOR encoding, reducing storage by **4.28×**.
- Benchmarked a **hash-indexed** and a **B+ tree index** at 1M entries — hash map wins lookups (500ns vs 3.8µs), B+ tree wins prefix scans by 3.4×, each losing where expected.
- Proved crash safety via **deterministic fault injection**, killing the process at 7 exact code points mid-write/compaction with **zero data loss**.

Low-Latency Matching Engine | *C++, CMake, GoogleTest, Hardware Performance Profiling* — GitHub

- Built a C++ **limit-order-book matching engine** with **price-time priority**, partial fills, and O(1) cancels.
- Benchmarked **four OrderBook data-structure designs** from a tree-based map to a cached flat-index array, **improving throughput 2.5–3.9× (up to 14.5M ops/sec).**
- Diagnosed a **10.3% regression** via hardware performance counters, traced to an uncached bitmap scan; fixed it with a **cached best-price index**, turning it into a **100.6% gain (up to 122% elsewhere).**

Relay | *Python, FastAPI, SQLAlchemy, PostgreSQL/pgvector, RAG, Redis/Celery* — Live Demo | GitHub

- Built an **LLM-orchestrated context engine automating cross-tool workflows** across GitHub, Slack, and Jira — 8 query modes helping teams find code, investigate issues, and capture engineering knowledge.
- Implemented **hybrid keyword/vector retrieval** and **relevance scoring**, separating true matches from noisy correlations across integrated systems.
- Built **asynchronous background jobs** with Celery and Redis for data ingestion and indexing, and a provider-agnostic LLM layer supporting OpenAI, Claude, Groq, and Gemini, backed by 300+ tests.

Technical Skills

Languages: C++, Python, SQL
Databases: PostgreSQL, MongoDB, Redis
Systems: System Design & Design Patterns, Multithreading, Distributed Systems, Performance Optimization, Testing
Backend & Infrastructure: FastAPI, REST APIs, Celery, SQLAlchemy
DevOps & CI/CD: Docker, AWS, Kubernetes, GitHub Actions (CI/CD), Shell Scripting, Prometheus, Grafana, SDLC
Developer Tools: Git, GitHub, Linux, Pytest, GoogleTest, Postman, Technical Documentation (ADRs)
AI & ML: LLMs, RAG, MCP, AI Agents, HuggingFace Transformers, NumPy, Pandas, scikit-learn

Achievements & Certifications

- Solved 300+ DSA problems in C++; peak LeetCode contest rating of 1862 (Knight).
- Pearson "Certified Kubernetes Application Developer (CKAD)" Specialization — [Coursera]